

		Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology Department: AI-ML/AI-RO/AI/AI-DS/CSE/MICRO/SAP/QUICK/ORACLE/IT/AERO		
Subject Name	PROBABILITY, STATISTICS AND NUMERICAL METHODS	A.Y	2025/2026	
Subject Code	303191251	Semester	4 th	
Tutorial-6				
Sr No	Question	COs	B.T	Competence
1	Use the Trapezoidal Rule to evaluate $\int_0^4 \frac{1}{1+x^2} dx$ using $h = 1$.	6	3	Apply
2	Use Simpson's 1/3 Rule to evaluate $\int_0^3 (x^2 + 1) dx$ using 6 subintervals.	6	3	Apply
3	Apply Simpson's 3/8 Rule to compute $\int_0^2 (e^x) dx$ using 6 subintervals.	6	4	Analyze
4	Evaluate $\int_0^1 \frac{1}{\sqrt{1+x}} dx$ using the Trapezoidal, Simpson's 1/3 and Simpson's 3/8 Rules and compare results.	6	5	Evaluate
5	Use the Taylor Series Method to approximate $y(0.1)$ for $dy/dx = x + y$, with $y(0) = 1$.	6	3	Apply
6	Using Euler's Method, compute $y(0.2)$ and $y(0.4)$ for $dy/dx = y - x$, with $y(0) = 1$ and $h = 0.2$.	6	4	Analyze
7	Apply Modified Euler's Method to find $y(0.1)$ for $dy/dx = x^2 + y^2$, with $y(0) = 0$ and $h = 0.1$.	6	4	Analyze
8	Use Runge-Kutta 4th Order Method to find $y(0.2)$ for $dy/dx = x + y$, with $y(0) = 1$ and $h = 0.1$.	6	6	Create